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Tutorial - 2

Q. Compare various SDLC models and justify the selection of an appropriate model for developing an online shopping system.

Introduction:

The Software Developement Life Cycle (SDLC) is a structured process used by software organizations to design, develop, and test high-quality software. It provides a systematic framework that defines tasks performed at each step in the software developement process. Selecting the appropriate SDLC model is critical as it determines the pace of developement, the handling of risks, the level of customer interaction, and ultimately the success or failure of the project.

Project overview: Pricepick.

Pricepick is an innovative web application designed to revolutionizing the online shopping experience. It combines AI-driven personalized product recommendations with a robust cross-site price comparison engine. Unlike traditional comparison table Pricepick compares and uses gemini api to understand user preferences and suggest products they are genuinely interested in, while simultaneously finding the best deals across multiple platforms.

Key Requirements:

AI-powered Recommendation :- Utilization of Collaborative and content-based filtering algorithms.

Real-time Data Aggregation :- Scraping and aggregating pricing data from platforms like Amazon, flipkart, etc.

Cross-platform comparison :- Normalising product data to allow direct feature and price comparison.

Technology Stack:

Frontend: React.Js with Tailwind Cs.

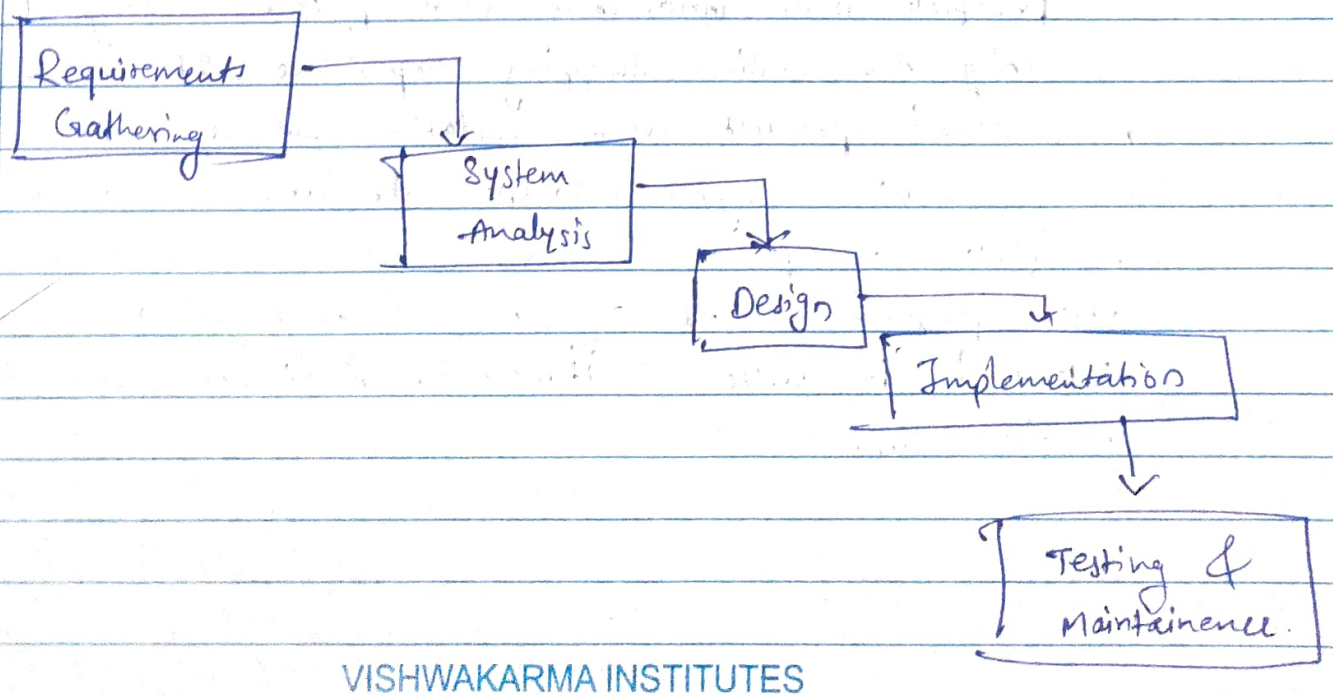
Backend: Node.Js (Express) & Python (flask / FastAPI for AI)

Infrastructure: Docker & Kubernetes.

SDLC Model Analysis :-

Waterfall model:

The Waterfall model is a linear sequential flow where progress flows steadily downwards through phases like a waterfall.



Spiral is too costly and complex for a student mid/sized project
Agile offers the best balance of flexibility, risk management and continuous delivery required for modern web application

Analysis and Justified Selection:

Price Pick project analysis:
Developing pricepick involves several complexity factors that influence the choice of SDLC.

Uncertainty in AI/ML Performance: The recomendation engine (collaborative engine) requires experimentation. We cannot define the "perfect" model upfront; it needs iterative training and tuning based on data.
External dependencies: The system relies on scraping or API integration with multiple ecomerce sites (Amazon Flipkart) These external platforms frequently change their structures, requiring our system to adapt quickly.
Real-time Requirement: Price comparison needs to happen in real time requiring efficient Backend optimization which might need multiple refactoring cycles.
Technology stack integration: Integrating distinct components like React Frontend, Node.Js API, Python ML service within Docker container requires a development process that supports modular integration and testing.

Justified Recommendation: Agile Method

V Model (verification and validation)

An extension of the waterfall model where process steps are bent upwards after coding phase. It demonstrates the relationship between each phase of the development life cycle and its associated phase of testing.

Requirements	Acceptance testing
System design	System Testing
Arch. design	Integration testing
Module design	Unit Testing
Coding	

Advantages: Proactive defect tracking - that is defects are found at an early stage

Disadvantages: very rigid and least flexible; software is developed during the implementation phase so, no early prototype are produced.

Suitable for: Projects where requirements are clearly defined and fixed (eg: medical device

suitable for : Projects with changing requirements, such as AI / web startups.

- Comprehensive Comparison of SDLC Models:

The following table compares the discussed models accross critical software engineering parameters to aid in selection.

Criteria	Waterfall	Incremental	Prototype	Spiral	V model	Agile.
flexibility	Low	moderate	High	moderate	Low	very High
Risk Mang	Low	moderate	moderate	HIGH	Low	High
Cost	Low	moderate	high	very high	Moderate	variable
Time to 1st Prototype	Late	early	earliest	early	Late	early
Customer Involvement	start only	Intermediate	high	high	start/ End	Continuous
Documentation	Heavy	Moderate	Low	Heavy	Heavy	Light
Testing Approach	End of cycle	every increment	Continuous	end of phase	Parallel to Dev	Continous
change Mgn	Difficult	easy	easy	Easy	Difficulty	very ea
Teamsize	Large	Large	small	large	Large	small/M
Suitable for	fixed req	SYS	UI intensive	high Risk	Critical sys	evolvin reqs.

Key Takeaways for Pria Pick:

waterfall and v model are too rigid for an AI project where model accuracy is unknown initially

they are genuinely intrested in, while simultaneously finding the best deals accross multiple platforms.

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SDLC Model analysis

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Risk Mitigation:

By developing working software every 2-3 weeks we mitigate the risk of total project failure. If the ML component is too difficult, we still have a working Price comparison tool ready to show.

Implementation Strategy

The development will follow a 2 week sprint agile. The team will hold:

- Daily standups: To discuss blockers or API issues.
- Sprint Planning: To select features from Product Backlog
- Sprint Review: To demonstrate the current features to guide

Conclusion:

for the Price Pick project, the rigid structure of waterfall or V-model would likely lead to failure due to dynamic nature of e-commerce data and AI-tuning. The Agile model provides the necessary framework to handle complexity through small, manageable iterations ensures the application stays adaptable to external changes, and guarantees that a working product is available for review throughout the semester.

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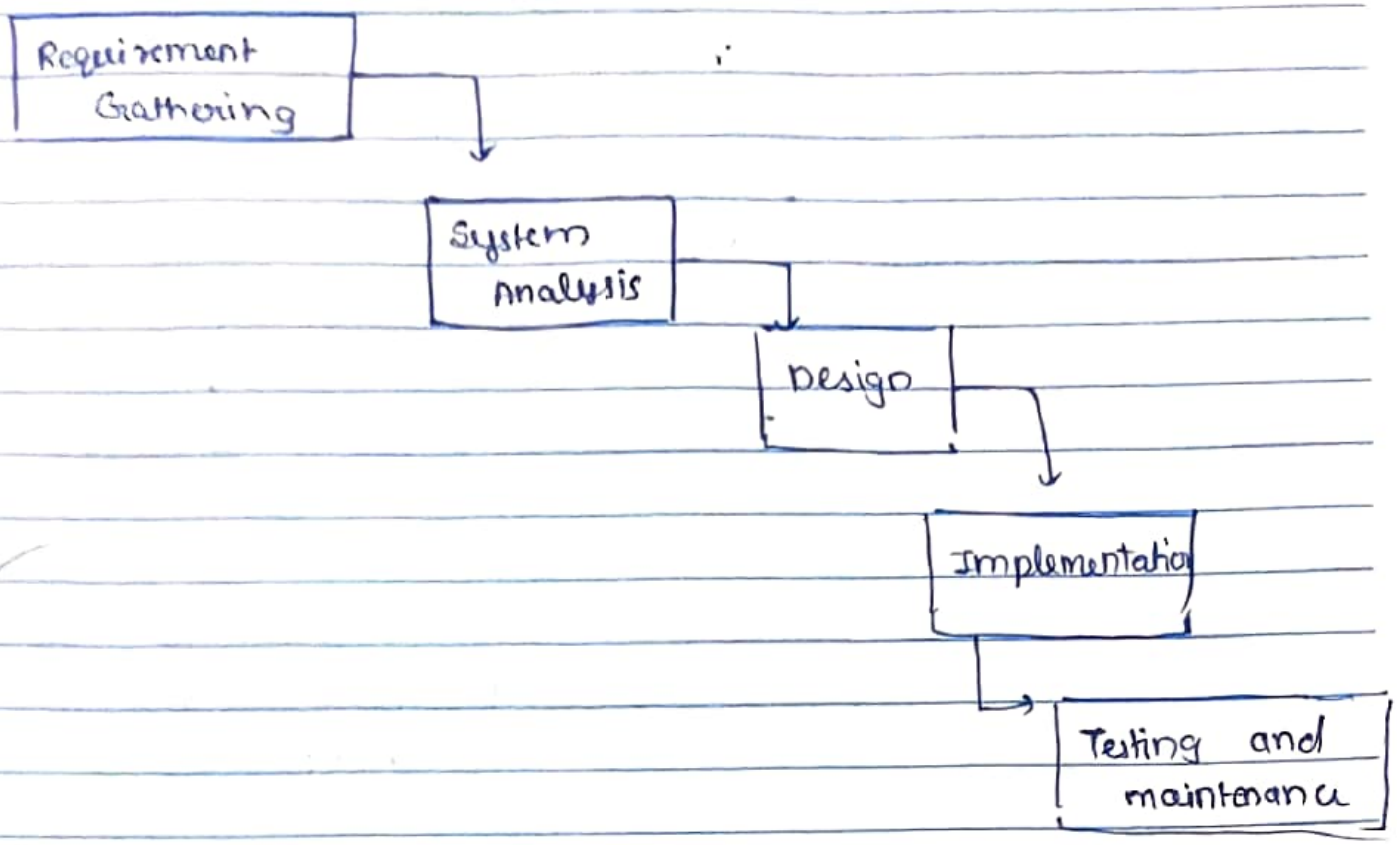


Advantages: Simple to understand, phases are processed and completed one at a time, clearly defined milestones.

Disadvantages: No working software is produced until late in the life cycle, high risk and uncertainity

Not suitable for complex object-oriented projects, difficulty to measure progress within stages.

Suitable for: small projects where requirements are very well understand and fixed.

Increment model:

The product is designed, implemented, integrated and tested as a series of incremental builds.

Advantages: Users are actively involved in the development; Errors can be detected much earlier.
Disadvantages: Practically, this methodology may increase the complexity of the system, incomplete application may cause focus on easy tasks first.
Suitable for: Systems with nuclear user requirements or complex UI interactions

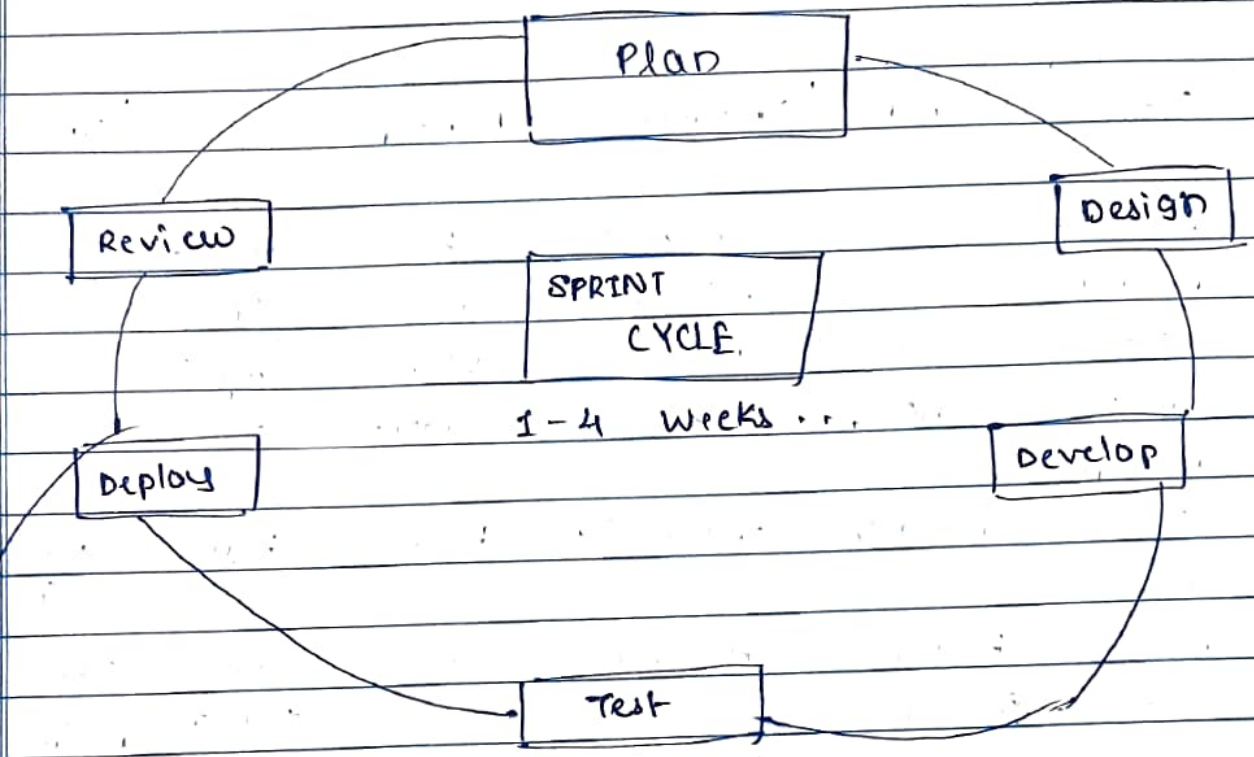
Spiral Model:
The spiral model combines the idea of iterative development with the systematic, controlled aspects of the waterfall model. It emphasizes Risk analysis.

Determine objectives	Identify & Resolve Risk	(∴ Spiral moves outwards through these 4 quadrants)
Plan Next phase	Develop & Test	

Advantages: High amount of risk analysis; Good for long and mission critical projects.
Disadvantages: Can be costly model to use; Risk analysis requires specific expertise
Suitable for: medium to high risk projects.

Agile Model.

Agile is a combination of iterative and incremental process models with focus on process adaptability and customer satisfaction by rapid delivery of working software products.



Advantages: Customer satisfaction by continuous delivery of useful software, people and interaction are emphasized rather than process and tools; working software is delivery frequently.

Disadvantages: Difficulty is assessing the effort required at the beginning of the software development life cycle; Documentation is often neglected.

Based on the comparative analysis and project constraints, the Agile methods (specifically Scrum framework) is selected as the most appropriate model for developing Price Pide.

Justification Points:

Iterative development for ML Refinement.
Machine learning models are rarely perfect in the first iteration. Agile allows us to deploy a basic version of the recommendation engine, gather user interaction data and refine the model in subsequent sprints.

Sprint Based Modular development:
The project can be broken down into manageable sprints.

3) Sprint 1: Basic UI and User authentication.
- Sprint 2: Data Aggregation (Scraper) for one site.
- Sprint 3: Price comparison logic
- Sprint 4: ML recommendation engine Integration:

4) DevOps and CI/CD Integration:
The project uses docker and Kubernetes. Agile pairs perfectly with devops practices, allowing for continuous integration Every time a feature is finished.

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V Model (Verification

Increment 1 Analysis → Design → Code → test	→ core features!

Increment 2 Analysis → design → code → test	→ Add'L Features

Increment 3: Analysis → design → code → test	→ Final product

Advantages: Generates working software quickly and earling ducing the life cycle, More flexible - less to change spope and requirment.

Disadvantages: Needs good planning and design; Total cost is higher than waterfall.

Suitable for: Projects where requirements are known but expected to evolve overtime.

Prototype model:
A prototype is built, tested and then reworked as necessary until an acceptable prototype is achieved from which the complete system or product can now be developed.

Listen to customer	→	Build Revise Mock up	→	Customer test-drive